

EMANUEL MILLER

Senior AI & Full Stack Engineer

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SUMMARY

Full-stack and applied AI engineer building applications across frontend, backend, and model integration. Focused on LLM applications, RAG, and AI agents, with a practical approach to architecture, evaluation, and reliable delivery.

EXPERIENCE

Senior AI & Full Stack Engineer

Ptolemy

06/2024 - 05/2026 Walnut, CA

- Developed AI-powered applications across frontend interfaces, backend services, and LLM integrations, translating business requirements into practical product features.
- Built RAG pipelines connecting document processing, semantic search, and source attribution.
- Developed agentic workflows using tool calling, structured outputs, and API integrations.
- Optimized retrieval, prompt construction, and backend processing for response quality, latency, and efficiency.
- Contributed to architecture reviews, automated testing, and production monitoring.

AI / Machine Learning Engineer

SolidBrain

07/2021 - 05/2023 Austin, TX

- Developed machine-learning workflows covering data preparation, experimentation, model evaluation, and application integration.
- Built reusable Python pipelines for data transformation and model development.
- Evaluated model performance using baseline comparisons and error analysis.
- Explored LLM and retrieval-based prototypes for document processing and knowledge retrieval.
- Collaborated with engineers and business stakeholders to integrate AI capabilities into application workflows.

TECHNICAL STACK

Full-Stack Development

React, Next.js, Node.js, Django, Python, JavaScript, HTML, CSS, REST APIs

AI Engineering

LLM Applications, RAG Pipelines, Semantic Search, Prompt Engineering, Structured Outputs

AI Agent Development

Tool Calling, API Integration, Context Management, Stateful Orchestration, Human-in-the-Loop Workflows, SOC2 Compliance

Model Development

Data Preprocessing, Feature Engineering, Model Evaluation, Error Analysis

Architecture & Reliability

Software Architecture, Modular Applications, Frontend/Backend Integration, Automated Testing, Code Review, Production Monitoring

KEY ACHIEVEMENTS

Legacy Modernization

Championed the modernization of legacy applications, improving overall system performance and reliability by **98%** while supporting higher data loads efficiently.

System Performance

Improved application performance by **up to 98%** through API query optimization, more efficient backend processing, and improved data retrieval.

EXPERIENCE

Senior Software Engineer

DataArt

📅 09/2018 - 02/2020 📍 New York, NY

- Developed full-stack application features and maintainable frontend and backend services.
- Designed and integrated APIs with validation, error handling, and database access.
- Improved performance through query optimization and efficient backend processing.
- Investigated production issues and implemented reliability improvements.
- Collaborated with product and QA teams on planning, code reviews, testing, and releases.

Full Stack Developer

XeoDev

📅 04/2014 - 02/2017 📍 Orlando, FL

- Developed and maintained web applications using React, Node.js, Python, and Django.
- Built responsive interfaces and reusable components.
- Implemented API endpoints supporting business logic, validation, and database operations.
- Improved frontend responsiveness and data retrieval efficiency.
- Worked with designers, developers, and testers on defects, browser compatibility, and releases.

EDUCATION

Bachelor of Science, Computer Science

University of Miami

📅 04/2001 - 2017 📍 Florida

Bachelor of Engineering, AI

Cornell University

📅 03/2008 - 09/2012 📍 New York

LANGUAGES

English

Native or bilingual proficiency